

BUBBLE BLASTER

Make the bubbles move

There are now lists to store the ID, size, and speed of the bubbles, which are randomly generated. The next stage is to write the code that makes the bubbles move across the screen.

5 This function will go through the list of bubbles and move each one in turn.

```
def move_bubbles():
    for i in range(len(bub_id)):
        c.move(bub_id[i], -bub_speed[i], 0)
```

Goes through each bubble in the list

Moves the bubble across the screen according to its speed

Imports the functions you need from the Time library

6 This will be the main loop for the game. It will be repeated over and over while the game is running. Try running it!



Don't forget to save your work

```
from time import sleep, time
```

```
BUB_CHANCE = 10
```

```
#MAIN GAME LOOP
```

```
while True:
```

```
    if randint(1, BUB_CHANCE) == 1:
```

```
        create_bubble()
```

```
    move_bubbles()
```

```
    window.update()
```

```
    sleep(0.01)
```

Generates a random number from 1 to 10

If the random number is 1, the program creates a new bubble (on average 1 in 10 times – so there aren't too many bubbles!)

Runs the "move_bubbles" function

Updates the window to redraw bubbles that have moved

Slows the game down so it's not too fast to play

7 Now you're going to create a useful function to find out where a particular bubble is, based on the ID. This code should be added to the program directly after the code you created in step 5.

```
def get_coords(id_num):
```

```
    pos = c.coords(id_num)
```

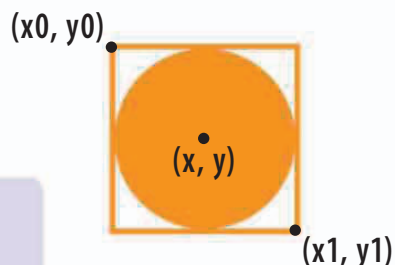
```
    x = (pos[0] + pos[2])/2
```

```
    y = (pos[1] + pos[3])/2
```

```
    return x, y
```

Works out the x coordinate of the middle of the bubble

Works out the y coordinate of the middle of the bubble



△ Locating bubbles

The function finds the middle of the bubble by taking the point halfway between the corners of the box around it.